

Skills Summary

Product and Quotient Laws for Exponents

Use this reference while completing your worksheet or when studying.

Skill 1 — The expanded-form model

What an exponent means: $x^4 = x \cdot x \cdot x \cdot x$ — the exponent is a *count of factors*, nothing more.

Why the laws work: multiplying two powers writes the two groups of factors end to end, so the counts add. Dividing cancels matching factors top and bottom, so the counts subtract. Expand whenever a law feels uncertain — the model never lies.

Example: Use expanded form on $2^2 \cdot 2^3$.

Step 1. The claim is about how many factors there are, so I write them all out and count rather than reaching for a rule.

Step 2. Two factors followed by three more makes a run of five factors of the base, and multiplying that run out gives the value.

Exponent = **5**, value = **32**.

Try it: Use expanded form on $3^2 \cdot 3^2$: give the exponent and the value.

check: exponent 4, value 81

Skill 2 — The product law

Law: $x^m \cdot x^n = x^{m+n}$ — same base, add the exponents.

It holds for negative exponents too: adding a negative count simply cancels factors. When numbers ride in front, multiply those separately: $(3x^2)(4x^5) = 12x^7$.

Example: Simplify $x^6 \cdot x^2$.

Step 1. Same base and a multiplication, so the product law applies and the exponents add.

Step 2. Six factors followed by two more is a single run of eight factors of the same base.

$\Rightarrow x^6 \cdot x^2 = x^8$

Try it: Simplify $w^3 \cdot w^7$.

check: w^{10}

(!) Watch out: multiplying powers adds the exponents; it never multiplies them. Test any rule you are unsure of on small numbers you can check by hand.

Skill 3 — The quotient law

Law: $\frac{x^m}{x^n} = x^{m-n}$ — same base, subtract the bottom exponent from the top one.

If the bottom exponent is larger, the result has a negative exponent; rewrite it as one over the positive power. Numbers in front divide separately.

Example: Simplify $\frac{k^9}{k^4}$.

Step 1. Same base and a division, so the quotient law applies and the exponents subtract.

Step 2. Four factors on the bottom cancel four of the nine on top, leaving the difference as the count.

$$\Rightarrow \frac{k^9}{k^4} = k^5$$

Try it: Simplify $\frac{p^5}{p^8}$, answer with a positive exponent.

check: $\frac{1}{p^3}$